

Supporting Information 1: Impact of filtering of protein/peptide lists on FDR estimates.

2024-11-26

Introduction

Some may assume that shortening the list of proteins would artificially increase the false discovery rate (FDR). However, the FDR, estimated using the Benjamini-Hochberg correction, depends primarily on the p-value distribution relative to the distribution of p-values under the null hypothesis (H_0 , a uniform distribution), not the absolute number of proteins tested. While methods controlling the Family-Wise Error Rate (FWER) are sensitive to the total number of tests, the FDR estimation is unaffected as long as the p-value distribution remains unchanged.

The computer Experiment

Example 1, filtering without changing the distribution of p-values

We start by simulating a list of $m = 3000$ p-values. 90% datapoints comes from H_0 , 10% from H_1 (fold change of 2). The group sizes are 4. We store these p-values in the array *pvals24*.

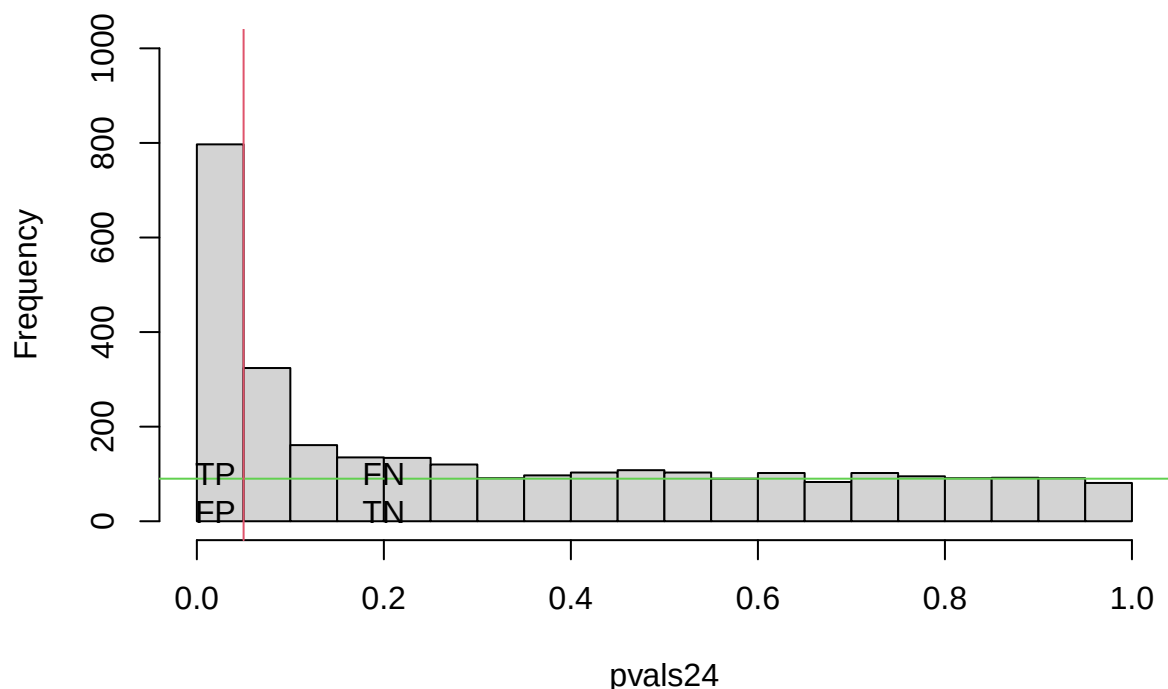
```
m <- 3000
simulate.p.values <- function(
  i, delta = 2, fraction = 0.1, ss = 4){
  control <- rnorm(ss,0,1)
  treatment <- rnorm(ss,0,1)
  if (runif(1) < fraction)
    treatment <- treatment + delta
  return(t.test(treatment,control)$p.value)
}
pvals24 <- sapply(1:m, simulate.p.values,
  delta = 2, fraction = 0.4 )
```

We plot the distribution of the p-values and see that we can estimate FP TP FN and TN from the histogram, which allows us to compute the false discovery proportion (FDP).

$$FDP = \frac{FP}{FP + TP}$$

```
hist(pvals24, breaks = 20, ylim = c(0,m/3), main = "B")
abline(h = (m - m * 0.4)/20, col = 3)
abline(v = 0.05, col = 2)
text(x = c(0.02, 0.02, 0.2, 0.2), y = c(20,100,20,100), labels = c("FP","TP","TN","FN"))
```

B



We can also compute the FDR using the Benjamini-Hochberg method using the function `p.adjust`.

```
xx <- data.frame(p.vals = pvals24, FDR = p.adjust(pvals24, method = "BH"))
```

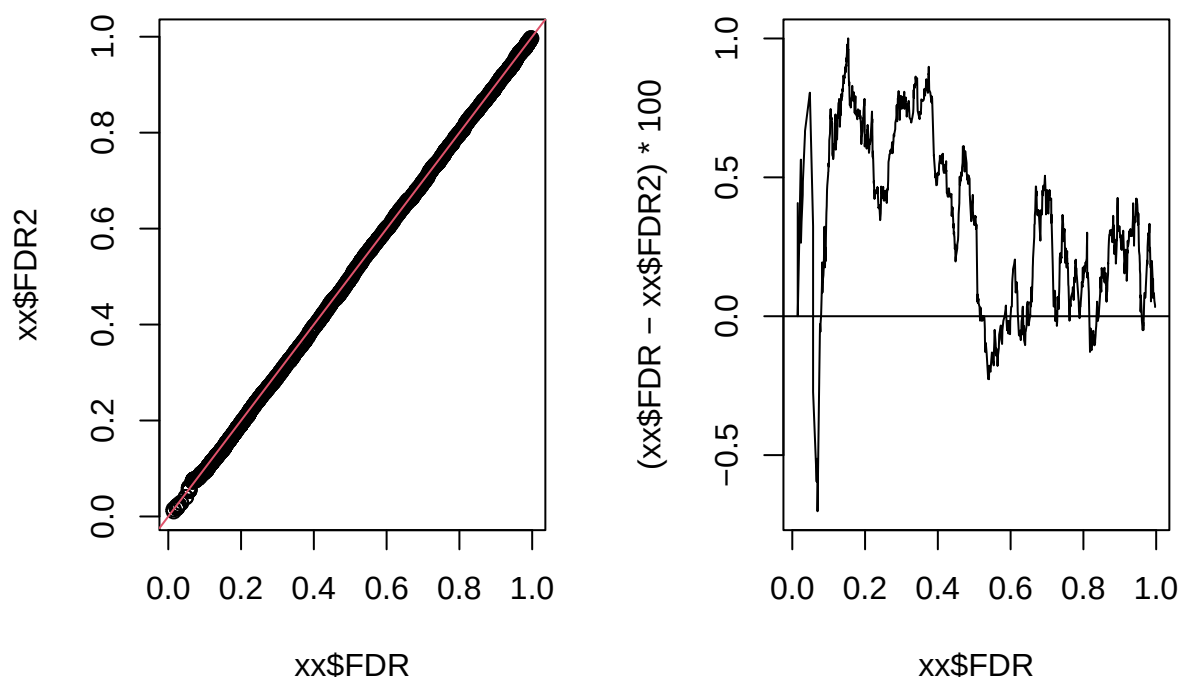
In proteomics experiments, only a single peptide identifies approximately 1/3 of proteins. If we remove these proteins from the list, and keep only those quantified by two or more peptides, do we need to recompute the FDR?

We filter the simulated p-values, and keep only 2/3 of them. We randomly choose which rows of data to use. We then compute the *FDR* from the remaining *p-values*.

```
xx <- xx[sample(1:m, floor(m*2/3)),]  
xx$FDR2 <- p.adjust(xx$p.vals, method = "BH")
```

We compare the FDR estimates computed for all 3000 proteins and the 2000 proteins left after filtering. We can see that the differences in the p-value estimates are less than 1.

```
par(mfrow = c(1,2))  
xx <- xx[order(xx$p.vals),]  
plot(xx$FDR, xx$FDR2)  
abline(c(0,1), col = 2)  
plot(xx$FDR, (xx$FDR - xx$FDR2)*100, pch = ".", type = "l")  
abline(h = 0)
```



We examine how this affects the list of selected proteins if we filter for FDR at 5%, 10%, or 25%.

```
l1 <- data.frame(FDR = c("5" = sum(xx$FDR < 0.05),
"10" = sum(xx$FDR < 0.1), "25" = sum(xx$FDR < 0.25)),

FDR2 = c("5" = sum(xx$FDR2 < 0.05), "10" = sum(xx$FDR2 < 0.1), "25" = sum(xx$FDR2 < 0.25)))
knitr::kable(l1)
```

	FDR	FDR2
5	16	16
10	235	265
25	723	735

We see that we end up with similar lists of proteins, and it does not matter if we used the *FDR* computed before or after removing 1/3 of observations.

Example 2, filtering and changing the distribution of p-values.

We now change the distribution of the p-values by filtering the list of proteins and then recomputing the FDR.

We augment the original list of 3000 p-values with 10 p-values equal to 1 and compute the FDR.

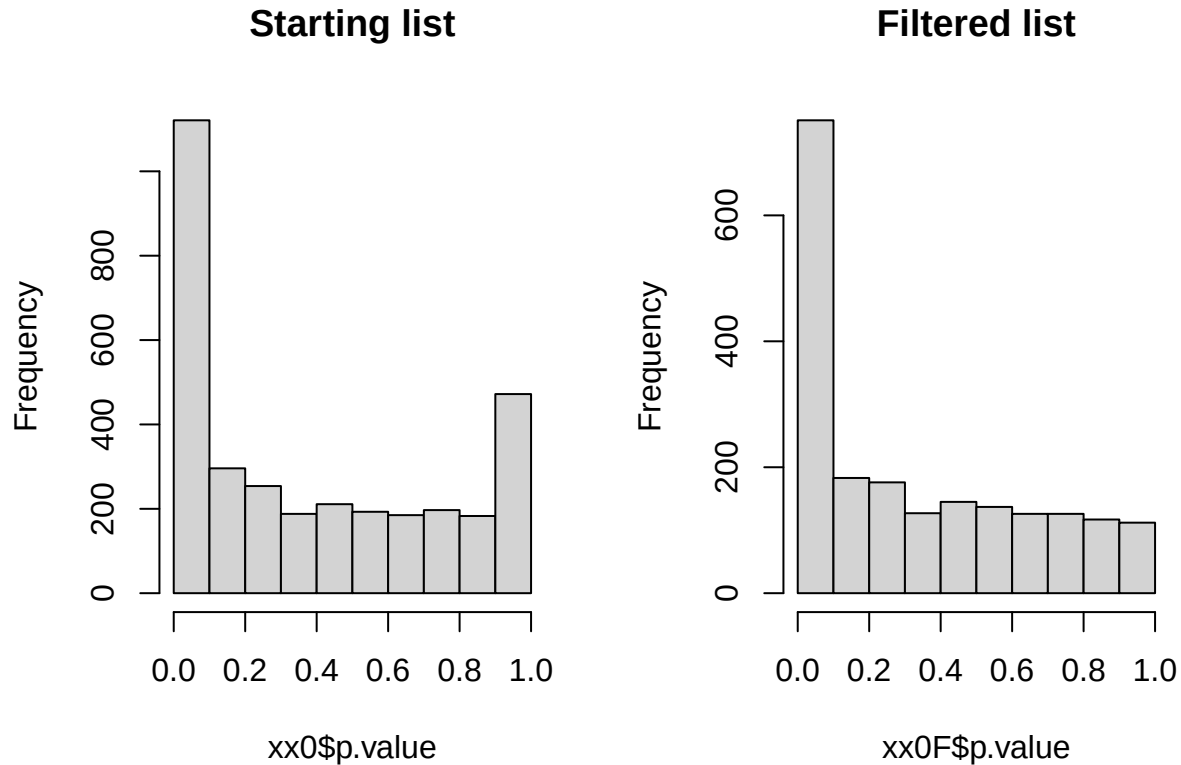
```
addP <- ceiling(m*0.1)
pvals240 <- c(pvals24, rep(1,addP))
xx0 <- data.frame(p.value = pvals240, FDR = p.adjust(pvals240, method = "BH"))
```

Next, we remove those p-values equal 1, and in addition 33% of the other p-values, leaving 2000 p-values.

```
xx0F <- xx0[-(3001:(3000 + addP)),]
xx0F <- xx0F[sample(1:m,floor(m*2/3)),]
```

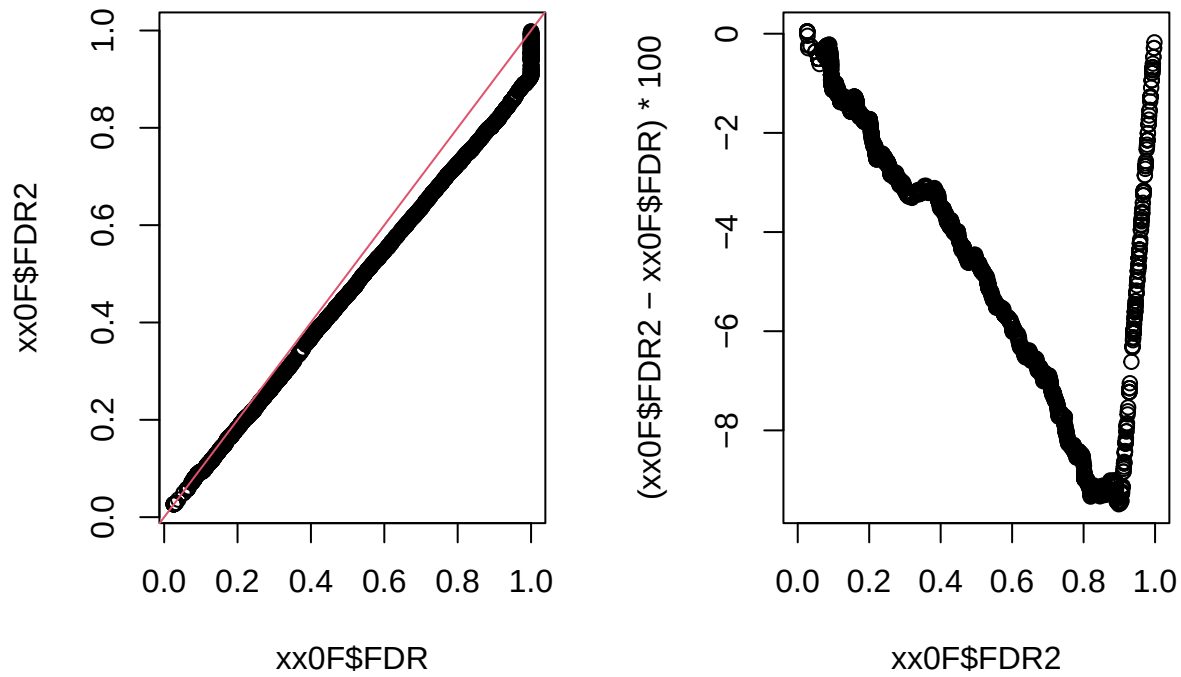
Clearly, the p-value distributions do differ before and after filtering.

```
par(mfrow = c(1,2))
hist(xx0$p.value, main = "Starting list")
hist(xx0F$p.value, main = "Filtered list")
```



Now we do recompute the FDR from the truncated list of p-values, and compare the FDR estimates computed before with those computed after filtering the lists.

```
xx0F$FDR2 <- p.adjust(xx0F$p.value, method = "BH")
par(mfrow = c(1,2))
plot(xx0F$FDR, xx0F$FDR2)
abline(0,1,col = 2)
plot(xx0F$FDR2, (xx0F$FDR2 - xx0F$FDR)*100, xlim = c(0,1))
```



The difference between the FDR computed before and after filtering is substantial. Again, we examine, how this affects the list of selected proteins if we filter for FDR at 5%, 10% or 25%.

```
l1 <- data.frame(FDR = c("5" = sum(xx0F$FDR < 0.05),
"10" = sum(xx0F$FDR < 0.1), "25" = sum(xx0F$FDR < 0.25)),
FDR2 = c("5" = sum(xx0F$FDR2 < 0.05), "10" = sum(xx0F$FDR2 < 0.1), "25" = sum(xx0F$FDR2 < 0.25)))
knitr::kable(l1)
```

	FDR	FDR2
5	11	13
10	137	223
25	671	713

The lists of proteins/peptides filtered by FDR at 5%, 10%, and 25 have different lengths and become relatively smaller when FDR thresholds are larger.